

Dev Shree Saini

DOB – 29.12.2001, Indian



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Geomatics Master's student at ETH Zurich, working on Earth observation, geospatial machine learning, and environmental monitoring. Experienced in satellite data processing, deep learning for remote sensing, and building Python-based spatial modelling pipelines. Interested in applying Earth observation and machine learning to real-world environmental problems, from flood monitoring to ecosystem mapping.

Education

• Masters in Geomatics Engineering, ETH Zurich

Zurich, Switzerland

GPA: 5.33/6.0

Sept 2024 - Mar 2027

Semester Project-1: Detection of Heavy Precipitation using GNSS Zenith Wet Delay

Semester Project-2: Better maps of plant functional traits - towards planttraits.earth v2

Relevant Coursework: Deep Learning, Introduction to Machine Learning, Multivariate Statistics and Machine Learning, Space Geodesy, Geospatial Research Methods

• BTech in Civil Engineering, VIT-Vellore

Vellore, India

CGPA: 9.33/10.0 – Ranked 2nd in department (~150 students), Best outgoing student

July 2019 - May 2023

Thesis topic: Flood Risk Analysis using Arc-SWAT in the Dhansiri River Watershed, India.

Scholarship: Merit-based scholarship for 3 years for academic excellence.

Relevant Coursework: Hydrology, GIS and Remote Sensing, Environmental Engineering

Professional Experience

Remote Sensing Laboratories, University of Zurich, Research Intern

Zurich, Switzerland

ESA-funded project on hyperspectral environmental monitoring:

July 2026 - Present

- Built an end-to-end processing pipeline for airborne hyperspectral and satellite imagery.
- Validated cross-sensor and multi-resolution scaling methods against real satellite data.
- Analysed how resolution and observing conditions affect vegetation trait retrieval for a mission study.

Key Skills: Python, Machine Learning Regression, Multi-sensor Data Processing, Scientific Writing

Department of Space Geodesy, ETH Zurich, Research Assistant

Zurich, Switzerland

Within the space geodesy research team:

Sept 2025 – Mar 2026

- Built Python pipelines to process multi-year GNSS and ERA5 reanalysis data (NetCDF/GRIB) for atmospheric analysis.
- Developed deep learning and gradient-boosting models to predict heavy rainfall from atmospheric water vapour.
- Scaled analysis from 1 to 12 years across ~5,500 European stations, including quality control and outlier detection.
- Presented these results at the [Swiss Geoscience Meeting](#), Bern, Dec 2025.

Key Skills: GNSS Meteorology, ERA5 Reanalysis, Time-Series Forecasting, Python, PyTorch

Indian Institute of Remote Sensing (IIRS), ISRO, Spatial Data Analyst Intern

Dehradun, India

As a part of the research group:

Jan 2023 – Dec 2023

- Developed a U-Net model in PyTorch on Sentinel-1 data for flood detection in monsoon-affected regions.
- Integrated four years of gridded rainfall data as additional model input channels.
- Tuned training pipelines and hyperparameters, reaching 92% accuracy and 0.70 F1 on held-out data.

Key Skills: Deep Learning, PyTorch, Model Evaluation, Geospatial Data Processing

Vellore Institute of Technology, Teaching Assistant

Vellore, India

Involved in many research projects and coursework in final year of bachelors:

Sept 2021 – Jan 2023

- Assisted in 3 PhD research projects involving geospatial and hydrological modelling; mentored ~40 students in QGIS and ArcGIS.

Key Skills: Teaching, Communication, Spatial Data Analysis

Projects and papers

Projects:

- [Semester Project-1](#): Detection of Heavy Precipitation using GNSS Zenith Wet Delay
 - Used GNSS Zenith Wet Delay as a physical proxy for atmospheric water vapour to predict short-term heavy rainfall.
 - Processed GNSS and ERA5 reanalysis data (NetCDF/GRIB) from ~3,200 European stations.
 - Benchmarked LSTM, XGBoost and further architectures under identical train/test splits.
 - LSTM performed best, reaching the highest F1 score (0.70) among all tested architectures.
- [MapSAM2](#): Adapting a transformer-based foundation model for automatic map segmentation
 - Adapted and fine-tuned the SAM2 foundation model (transformer architecture) for segmentation of historical maps.
 - Implemented video-based tile processing to share spatial context across image patches.
- [Semester Project-2](#): Better maps of plant functional traits - towards planttraits.earth v2
 - Extended the [planttraits.earth](#) framework for global mapping of 37 plant functional traits.
 - Developed CNN, Multi-Task Learning, and Mixture-of-Experts models for multi-trait prediction.
 - Integrated 150-channel Earth Observation data with GBIF and sPlot biodiversity records.
 - Generated and evaluated global trait maps at 1 km and 22 km resolutions.

Research Papers (2 first-author publications in peer-reviewed journals):

- X. Xia., **Saini, D. S** et al. (2026). [MapSAM2: Adapting SAM2 for Automatic Segmentation of Historical Map Images and Time Series] ([10.1109/TGRS.2026.3678420](#)). IEEE Transactions on Geoscience and Remote Sensing.
- **Saini, D.S.** et al. (2025). [Land use/cover dynamics and associated impacts on eutrophication, land surface temperature, and ecosystem service values] ([10.1007/s12237-025-01509-0](#)). Estuaries and Coasts.
- **Saini, D. S.** et al. (2024). [Simulation of the hydraulic model HEC-RAS coupled with GIS and remote sensing to study the effect of river cross-section width in detecting flood-prone areas] ([10.17491/jgsi/2024/173843](#)). Journal of the Geological Society of India.
- Ghosh, S., **Saini, D. S.** et al. (2023). [The impact of short-duration precipitation events over the historic Cauvery Basin: A study on altered water resource patterns and associated threats] ([10.1038/s41598-023-41417-6](#)). Scientific Reports.
- Antonio C. Pereira, **Dev S. Saini** et al. [Wetland Discharge Water Filtration of a Mesotrophic Lake] [CSCE Annual Conference 2023](#), Moncton, Canada

Skills

- **Weather and Earth Observation Data**: ERA5 reanalysis, NetCDF, GRIB, GNSS, Sentinel-1/2, EnMAP
- **Programming and ML**: Python (PyTorch, NumPy, Pandas, xarray, GeoPandas, Rasterio), XGBoost, Git
- **Analytical**: Time-Series Forecasting, Deep Learning, Model Benchmarking and Evaluation, Large-Scale Data Pipelines, Atmospheric Data Analysis

Languages, and Interests

- **Languages**: Hindi (Native), English - C1 (IELTS), German - A2 (currently learning)
- **Interests**: Hindi Poetry, Chess, Cooking

Honors and Awards

- **Honors**: [MITACS scholarship](#), [Hackathon winner](#), [Scholarship Awards](#), [Rank Certificate](#), [All India Course Topper](#)

Volunteering and Positions

- Academic Head of an educational startup Mar 2020 – Mar 2022
- Core member of the university chapter ASCE Sept 2019 – Dec 2022